Avanthika Shydh

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Summary

AI/ML enthusiast and dual-degree student with hands-on experience in developing AI-driven solutions, including Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) systems, intelligent search applications, and predictive analytics. Skilled in Python, machine learning, data analytics, and computer vision, with exposure to AUTOSAR, Cloud Computing, Soft Computing, and Big Data Analytics. Passionate about leveraging AI and data-driven technologies to build intelligent systems for real-world applications in mobility, automation, and advanced engineering domains.

Work Experience

AI Intern | Tech Mahindra — April 2026 – May 2026

• Developed and enhanced AI-driven solutions as part of a Smart Laboratory Management System project, improving system functionality and user experience.

• Designed and implemented a regex-based search engine to enable efficient information retrieval and document search capabilities.

• Developed theoretical understanding of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, and AI application architectures.

• Contributed to system optimization, feature enhancement, and integration of intelligent automation components within enterprise software solutions.

• Collaborated with industry mentors and engineering teams, applying machine learning and software development principles to solve real-world business problems.

Education

Amrita School of Engineering Coimbatore, **B.Tech in Electrical and Computer Engineering** Aug 2023 – Present

- GPA: 8.04/10.0

IIT Madras , BS in Data Science and Applications Sept 2023 - Present

- Diploma Level Ongoing

St.Antonys Public School, Anakkal Mar 2021 – Mar 2023

- Score: 92.6%

SSVM Institutions ,Alangombu Jun 2009 – Mar 2021

- Score:98.8%

Projects

AI-Driven Traffic Monitoring System for Real-Time Congestion Detection and Route Optimization

• Engineered an AI-powered intelligent traffic system by developing advanced Computer Vision pipelines and LiDAR sensor fusion using Deep Learning models within CARLA simulation for real-time congestion prediction and adaptive signal control. Implemented proactive voice and SMS alert mechanisms, advancing safe, efficient, and autonomous urban mobility.

Published as a research paper at the ICISCoIS 2026, PSG College of Technology.

Crime Type prediction using Random Classifier

• Built a machine learning classification model to predict the type of crime most likely to occur in Indian districts. Worked with a dataset of 9000 rows and 29 features, applying extensive data preprocessing, feature engineering, district-wise categorical encoding, and hyperparameter tuning. Achieved 95% accuracy, evaluated with accuracy, precision, recall, and F1-score metrics.

• **Technologies:** Python, Pandas, Numpy, Scikit-learn, Excel, Random Forest Classifier, Data Cleaning and Feature Engineering, Model Evaluation

Face Authentication Enabled Hall Booking with Automated Power Control System

• Designed a smart college hall booking solution using face recognition for access validation, paired with microcontroller-enabled automated power control. Enhanced security, improved operational efficiency, and achieved significant energy savings during idle hours.

• **Technologies:** Raspberry Pi, Python (OpenCV), Flask, MySQL, HTML/CSS/JS, Relay-Based Power Automation

Smart Lab Management and Robotic-Arm Simulation System

• Engineered a scalable backend orchestration platform for autonomous chemical storage, retrieval, and robotic task execution using FastAPI and relational database architecture. Implemented workflow orchestration, task lifecycle management, real-time execution simulation, and 3D rack visualization to enhance operational reliability, traceability, and intelligent resource coordination.

• **Technologies:** FastAPI, Python, SQL, MySQL, REST APIs, Workflow Simulation, Backend System Design, 3D Visualization

Skills

Programming: Python, C, SQL, JavaScript, MATLAB,

Data Science & AI : Machine Learning, Deep Learning, Computer Vision, Feature Engineering, ETL, Data Cleaning, Data Modelling

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, OpenCV, YOLOv8

Analytics & Visualization: Power BI, Excel, Cognos

Systems & Tools: Git, GitHub, Jupyter Notebook, CARLA Simulator, Kaggle

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Computer Architecture, Embedded Systems, IoT, Statistics, Machine Learning, Deep Learning, LLMs, RAG

Certifications

IBM Data Analyst Professional Certificate
Developing Serverless Solutions on AWS